Direct count comparison with the physical variance allocation

Research note, 28 October 2026

The difficult negative phases can be handled without reducing the conditional support below 3/5. The useful information is asymmetric: a negative noise variable can make a large negative jump, whereas a positive noise variable has negative jumps of size at most 9/125. We retain this distinction for the first incoming count, and retain the same conditional third moments in the outgoing count and in the cubic penalty. A small closed certificate completes the resulting elementary comparison. The certificate has 291 cells; it does not discretize the sample size or the marginal distribution.

Theorem 1. *Let $n \ge 42$, $2/5 \le p \le 53/100$, $q = 1-p$, and $Y = B + U$, where B is Bernoulli with parameter p and $\mathbb{E}(U \mid B) = 0$. Write U_j for the conditional noise given $B = j$, and assume*

$$U_0 \in [-A, a_0], \qquad U_1 \in [-a_1, A], \qquad A = \frac{3}{5}, \quad a_0, a_1 \ge 0, \quad a_0 + a_1 \le d = \frac{9}{125}.$$

Set $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $\ell = (v+s)^{-1/2}$, $X = \ell(Y-p)$, $\rho = \mathbb{E}|X|^3$ and $\rho_B = (1-2v)/\sqrt{v}$. Let S_n be the sum of n independent copies of X. Suppose

$$\frac{3}{100} \le \lambda \le \frac{3}{40}, \qquad |H| \le \frac{46}{25}, \qquad np + H/\ell = k - r, \quad k \in \mathbb{Z}, \quad 0 \le r \le \frac{3}{20}.$$

For every $C \ge 2/5$, with $\delta = k - np$ and F_B the distribution function of $\mathrm{Bin}(n,p)$,

$$\begin{aligned}\Delta_C = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < 0.\end{aligned}$$

This statement concerns actual iid laws, including atom collisions and zero variance in either conditional law. It does not assume a least counterexample, contact, harmonic reserves, half-amplitude confinement, or marked crossings. Other phases and smaller noise require their separate comparison theorems. The present assertion alone is not a proof of an unrestricted Berry–Esseen bound, nor does it establish the acquisition of its strip hypotheses.

1 The count identity and its normalization

Write $v_j = \mathbb{E}U_j^2$, $\eta = 1/n$, and introduce the *physical* allocation

$$w = \frac{pv_1}{s}, \qquad \Lambda_0 = nqv_0 = \lambda(1-w), \qquad \Lambda_1 = npv_1 = \lambda w.$$

Let W_j be a sum of j independent variables with law U_1 and $n-j$ with law U_0, and put $\pi_j = \binom{n}{j} p^j q^{n-j}$. With $u = -r$, conditioning gives exactly

$$\begin{aligned}\mathbb{P}(S_n \le H) - F_B(k) = & -\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u + i) \\ & + \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u - i).\end{aligned}$$

The outgoing event is strict and the incoming event closed. Put

$$m = \sqrt{n}\pi_k, \qquad f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

If $\pi_{k+i}/\pi_k \le R$ and the sum of incoming probabilities is at most $\mathcal{T}$, it is sufficient to prove positivity of

$$m\{\mathbb{P}(W_k > -r) + f(-r) - R\mathcal{T}\} + C(\rho - \rho_B) - \frac{7}{4}(\Xi + r)\lambda\eta, \qquad \Xi = \frac{231}{250}. \tag{1}$$

Indeed the final term bounds the unfavorable normal variance correction: differentiating the normal distribution function gives at most $(1/5)v^{-3/2}\delta_+ s < (7/4)(\Xi + r)s$.

Here and below $v_* := 6/25 \le v$, $\eta \le 1/42$ and

$$\sqrt{v+s} < \frac{251}{500}, \quad \xi := H\sqrt{v+s}, \quad |\xi| < \Xi, \quad \delta = \xi + r.$$

In particular $1 \le k \le n-1$ and $9/25 < k/n < 23/40$. The finite Stirling estimate, with Taylor expansion around p, gives

$$m = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + e\}, \qquad n|e| \le Z(D) := \frac{5D}{8} + \frac{3}{8} + \frac{D^3}{42} \quad (|\delta| \le D \le \Xi + 3/20). \tag{2}$$

For clarity, the constants are respectively the prefactor error, factorial remainder, and entropy remainder. On the segment joining p to k/n, $|D'''(z\|p)| \le 6$, $|(\log(z(1-z)))'| \le 5/4$, and $1/(12z(1-z)) < 3/8$. Applying $0 < \log(j!) - (j+1/2)\log j + j - \frac{1}{2}\log(2\pi) < 1/(12j)$ gives the errors D^3/n^2, $5D/(8n)$ and $3/(8n)$. A proof of this finite factorial formula is retained in `manuscript/arxiv-v1/appendices/count-comparison.tex` (the paragraph beginning "For completeness, the finite Stirling formula used here is"). No asymptotic remainder in the noise is used.

On a closed cell $[l, L] \times [r_-, r_+] \times [w_-, w_+]$, let $D = \Xi + r_+$. Using $19943/50000 < \phi(0) < 2/5$ and $e^{-x} \ge 1 - x$ gives

$$m_- = 2\frac{19943}{50000}\left[1 - \frac{D^2/(2v_*) + Z(D)}{42}\right] \le m \le \frac{17}{20} =: m_+, \tag{3}$$

$$R = 1 + \frac{\max(0, 53/100 + D - 1)}{42v_* - (3/5)\max(0, D-1)}. \tag{4}$$

For the upper mass bound apply (2) at a mode, whose displacement from np is at most one, and use $e^{Z(1)/42} \le (1 - Z(1)/42)^{-1}$. For (4), the first exact ratio is $1 + (p - \delta - 1)/(nv + q(\delta+1))$; all subsequent ratios are at most one because $53/100 + D < 2$. All denominators and the bounds in this paragraph are checked rationally.

For fixed count and fixed δ, the derivative of $\log f'(t)$ with respect to s is

$$\frac{(\delta+t)^2 - n(v+s)}{2n(v+s)^2} < 0, \qquad -r \le t \le 0.$$

Indeed $|\delta + t| \le \Xi + r$ and $42v_* > (\Xi + 3/20)^2$. At $s = 0$, (2) therefore gives

$$\log f'(t) \le \eta P(r), \qquad P(r) = \frac{2\Xi r + r^2}{2v_*} + Z(\Xi + r).$$

Let $E = 42(\exp^+(P(r_+)/42) - 1)$, where $\exp^+$ is the rational upper enclosure described below. Convexity of the exponential on $0 \le \eta \le 1/42$ implies

$$f(-r) \ge -r(1 + E\eta). \tag{5}$$

This keeps the full $1/n$ scaling instead of introducing a constant error.

2 The first incoming count remembers the allocation

We use the quadratic-hinge comparison of Pinelis, *On the Bennett–Hoeffding inequality*, arXiv:0902.4058, Proposition 2.8 and Corollary 2.11. Independent centered variables bounded above by $b > 0$, with total variance at most V, have sum dominated in every quadratic hinge $(x-y)_+^2$ by $b(P_{V/b^2} - V/b^2)$. It applies with an upper bound on variance, not only its exact value. Conditioning on independent added variables preserves the comparison. Consequently the two groups in $-W_j$ may be replaced successively by independent centered Poisson variables with jump sizes A and d.

Lemma 1. *For fixed y, the function*

$$G(T,w;y) = \mathbb{E}\left[A\left(P_{T(1-w)/A^2} - \frac{T(1-w)}{A^2}\right) + d\left(P_{Tw/d^2} - \frac{Tw}{d^2}\right) - y\right]_+^2$$

is nondecreasing in $T \geq 0$ and nonincreasing in $0 \leq w \leq 1$.

Proof. Increasing T adds independent centered Poisson increments, so conditional Jensen applies. To compare jump sizes at fixed variance V, approximate $d(P_{V/d^2} - V/d^2)$ by a sum of N centered d-Bernoulli variables, with success probability $V/(Nd^2)$, for N sufficiently large. Their total variance is at most V and their upper bound is at most A. The quoted comparison bounds each quadratic hinge by the corresponding centered A-Poisson hinge. Their fourth moments are at most $d^2V + 3V^2$, giving uniform integrability of each quadratic hinge, so the inequality passes to the Poisson limit. It is preserved by adding an independent variable: condition first; a bound by a constant times $V + (Z-y)^2$ justifies integration for finite second moments. For $w_1 < w_2$, split the two comparators into common large-jump variance $T(1-w_2)$, common small-jump variance Tw_1, and the transferred variance $T(w_2 - w_1)$. Replacing the latter small jumps by large jumps gives the asserted direction. Zero variances are direct. □

The exact own-count variances are

$$T_0 = g_0\Lambda_0, \quad T_1 = g_1\Lambda_1, \qquad g_0 = 1 - \delta\eta/q, \quad g_1 = 1 + \delta\eta/p. \tag{6}$$

For an incoming count, $\delta + i = \xi + r + i > 0$. Thus its negative group has variance at most Λ_0, while its positive group has variance at most $\Lambda_1[1 + (5/2)(\Xi + r + i)\eta]$. The inflated comparator at the first incoming count therefore has total variance at most

$$T^+ = L\left[1 + \frac{5}{84}(\Xi + r_+ + 1)w_+\right], \tag{7}$$

and its fraction assigned to small jumps is at least the physical w. For any rational $0 \leq y < 1 + r_-$, Lemma 1 and the pointwise hinge inequality give the whole-cell bound

$$\mathbb{P}(W_{k+1} \leq -r - 1) \leq \frac{G(T^+, w_-; y)}{(1 + r_- - y)^2} =: Q. \tag{8}$$

One may replace Q by $\min(Q, 1)$.

The centered hinge is *not* assumed convex in w. Neither do we replace the own-count allocation by the physical allocation. In general $T_0/(T_0 + T_1) \neq 1 - w$; the exact factors in (6) must remain in the outgoing moment bound.

An exact finite expression, with no missing Poisson tail

Put $h = 3/125$, so $A = 25h$ and $d = 3h$. If $\mu = T(1-w)/A^2$, $\nu = Tw/d^2$, and $K = y/h + 25\mu + 3\nu \geq 0$, then $G = h^2 N$, where

$$N = 625\mu + 9\nu + (25\mu + 3\nu - K)^2 - e^{-\mu-\nu} \sum_{\substack{j,k\geq 0 \\ 25j+3k\leq K}} (K - 25j - 3k)^2 \frac{\mu^j \nu^k}{j!k!}. \tag{9}$$

This subtracts the lower complement from the full second moment. The sum is finite even for rational K, while the identity includes the complete upper tail. Its coefficients are nonnegative, so a lower exponential bound produces an upper bound for N. For $x \geq 0$, reduce to $z = x/2^a \leq 1$, sum the first 25 Taylor terms, and bound the remainder after order 24 by

$$\frac{z^{25}/25!}{1 - z/26}.$$

Repeated squaring gives rational endpoints for e^x; reciprocals give endpoints for e^{-x}. All arithmetic in acceptance uses integers and `Fraction`; no floating-point evaluation enters these bounds.

All later incoming counts

Every legal conditional count has total variance at most $n \max(v_0, v_1) \leq (5/2)\lambda$. For $i = 2, 3$ use

$$V_i = \min\left(\frac{5}{2}L, L\left[1 + \frac{5}{84}(\Xi + r_+ + i)\right]\right), \qquad \mu_i = V_i/A^2.$$

For an integer $j_i \geq 2$ with $(i + r_-)/A + \mu_i - j_i > 0$, the quadratic comparison and Poisson factorial moments give

$$\mathbb{P}(W_{k+i} \leq -r - i) \leq \frac{\mu_i^{j_i+1}}{(j_i+1)!((i + r_-)/A + \mu_i - j_i)^2} =: Q_i. \tag{10}$$

Indeed $(P - j)_+^2 \leq \binom{P}{j+1}$ for integers $P \geq 0, j \geq 2$. For $j = 2$ this is $(t-1)(t-2) \geq 0$ for integer $t = P - j \geq 1$; the binomial coefficient increases with j. Its Poisson expectation is $\mu^{j+1}/(j+1)!$. For $i \geq 4$, use the common variance $(5/2)L$, put $\mu_0 = (5/2)L/A^2$, and choose hinge $j = i$. The denominator increases with i, and the numerator ratio is at most $\mu_0/6 < 1$. Hence the entire remainder is

$$J = \frac{\mu_0^5}{5!\{4(1/A - 1) + r_-/A + \mu_0\}^2(1 - \mu_0/6)}. \tag{11}$$

We take $\mathcal{T} = Q + Q_2 + Q_3 + J$. Adding terms for nonexistent counts can only increase this upper bound. All unsquared hinge denominators are checked positive before division.

3 The cubic penalty and the outgoing count use the same moments

The signs of $Y - p$ on the two conditional supports are fixed, since $a_0 < p$ and $a_1 < q$. Put $M_0 = nq\mathbb{E}U_0^3$, $M_1 = np\mathbb{E}U_1^3$. Expansion gives

$$n\{\mathbb{E}|Y - p|^3 - v(1 - 2v)\} = 3p\Lambda_0 + 3q\Lambda_1 + M_1 - M_0.$$

The elementary inequality $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ implies

$$n(v+s)^{3/2}(\rho - \rho_B) \ge 3p\Lambda_0 + 3q\Lambda_1 + M_1 - M_0 - \frac{39}{50}\lambda - \frac{13}{16}\lambda^2\eta. \tag{12}$$

Here $(3/2)(1-2v) \le 39/50$ and $3(1-2v)/(8v) \le 13/16$. The right side is positive: it is at least $\lambda[6/5 - d - 39/50 - (13/16)\lambda\eta] > 0$. Therefore, for $K_0 = 79/25$, we may use $C/(v+s)^{3/2} > K_0$ in (12) *before* separating its signed terms. Also the uniform weaker consequence is

$$C(\rho - \rho_B) > \frac{(2/5)\{87/250 - (13/16)\lambda\eta\}}{(251/500)^3}\lambda\eta > \lambda\eta. \tag{13}$$

For $\alpha > 0$ define the quartic minorant

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2x^2 + 2\alpha^3x}{5\alpha^4} \le \mathbf{1}_{(0,\infty)}(x).$$

Its numerator is $x(x+\alpha)^2(2\alpha - x)$; its positive maximum is $(9+6\sqrt{3})\alpha^4/4 < 5\alpha^4$ because $108 < 121$. For a centered independent sum, expansion gives

$$\mathbb{E}W_k^4 = \sum \mathbb{E}Z_j^4 + 3t^2 - 3\sum(\mathbb{E}Z_j^2)^2, \quad \mathbb{E}W_k^3 = \sum \mathbb{E}Z_j^3, \quad t = T_0 + T_1.$$

Apply the minorant to $W_k + r$. To combine it with (12), price each conditional squared variable on its support. For group zero the price is

$$\frac{m}{5\alpha^4}g_0(x^2 + 4rx) + K_0\eta x, \qquad -A \le x \le a_0 \le d;$$

for group one it is

$$\frac{m}{5\alpha^4}g_1(x^2 + 4rx) - K_0\eta x, \qquad -d \le -a_1 \le x \le A.$$

Both are convex in x, because $m, g_0, g_1 > 0$. Thus their maxima are at the respective relaxed support endpoints $x_0 \in \{-A, d\}$ and $x_1 \in \{-d, A\}$. Dropping the favorable term $3\sum(\mathbb{E}Z_j^2)^2$ and replacing $3p \ge 6/5$, $3q \ge 141/100$ gives the following lower bound for (1), for the worst choice of x_0, x_1:

$$\begin{aligned}\mathcal{L} = \frac{m}{5\alpha^4}\big[&(3\alpha^2 - 6r^2 - x_0^2 - 4rx_0)T_0 + (3\alpha^2 - 6r^2 - x_1^2 - 4rx_1)T_1 \\ &\qquad - 3(T_0+T_1)^2 - r^4 + 3\alpha^2r^2 + 2\alpha^3r\big] \\ &- mr(1+E\eta) - mR\mathcal{T} - \frac{7}{4}(\Xi + r)\lambda\eta \\ &+ K_0\eta\lambda\left[\frac{6}{5}(1-w) + \frac{141}{100}w - x_0(1-w) + x_1w - \frac{39}{50} - \frac{13}{16}\lambda\eta\right].\end{aligned} \tag{14}$$

This is a same-law moment inequality. The cubic penalty is not allocated independently of the outgoing fourth-moment estimate.

4 Why the finite check covers every law and every sample size

For fixed r, the actual multipliers satisfy

$$1 - \frac{5}{2}(\Xi + r)\eta \le g_j \le 1 + \frac{5}{2}(\Xi + r)\eta.$$

The expression (14) is separately concave in g_0, g_1, so it suffices to substitute

$$g_j = 1 + \varepsilon_j\frac{5}{2}(\Xi + r)\eta, \qquad \varepsilon_j \in \{-1, 1\}.$$

This is a valid outer rectangle, even though its four corners need not be realized by one marginal. After substitution it is separately concave in each of λ, w, η: all nonaffine terms in any one of those variables are negative squares. It is affine in m. Hence it suffices to check $\lambda \in \{l, L\}$, $w \in \{w_-, w_+\}$, $\eta \in \{0, 1/42\}$ and $m \in \{m_-, m_+\}$, together with the four support-endpoint choices. Each choice leaves a polynomial of degree at most four in r. The artificial face $\eta = 0$ is an endpoint of a concavity argument, not a limit theorem for a fixed law.

For a polynomial $\sum_{k=0}^{4} c_k r^k$, its degree-four Bernstein coefficients on $[a, b]$ are

$$\beta_i = \sum_{k=0}^{4} c_k \sum_{j=0}^{\min(i,k)} \frac{\binom{k}{j}\binom{i}{j}}{\binom{4}{j}} a^{k-j}(b-a)^j, \qquad 0 \le i \le 4.$$

This follows by first substituting $r = a+(b-a)t$ and then using $t^j = \sum_{i=j}^{4} \binom{i}{j}\binom{4}{j}^{-1}\binom{4}{i} t^i (1-t)^{4-i}$. Positive coefficients prove positivity on the closed interval. There are 256 endpoint combinations per quartic cell and five coefficients per combination.

Some cells use the simpler Cantelli bound. Set

$$T = L[1 + (5/2)D/42], \quad h_f = 1 + E/42, \quad B = \min_{r \in \{r_-, r_+\}} \left\{ \frac{r^2}{T + r^2} - h_f r \right\} - R\mathcal{T}.$$

Require $r_- > 0$ and $2r_- T/(T+r_-^2)^2 > h_f$. The derivative of $r^2/(T+r^2) - h_f r$ first increases and then decreases, because its derivative has the sign of $T - 3r^2$. Under the displayed left-endpoint condition any interior critical point is a maximum, so the minimum is at an endpoint. The variance upper bound makes Cantelli uniform. Using (13), a valid margin is

$$\begin{cases} m_- B, & B \ge 0, \\ m_+ B, & B < 0 \end{cases} - \{(7/4)D - 1\}L/42. \tag{15}$$

The coefficient $(7/4)D - 1$ is positive on the domain.

The certificate `compound-certificate.json` contains 291 rational closed boxes, 200 using (14) and 91 using (15). Its domain is exactly

$$[3/100, 3/40] \times [0, 3/20] \times [0, 1].$$

The initial division has 3 noise intervals, 12 phase intervals and 8 allocation intervals. Three cells were split once, giving 291 final cells. The acceptance checker independently forms the common endpoint refinement (480 elementary boxes), checks each interior is covered exactly once and all exterior faces occur, and uses closed inequalities on every cell. Missing, duplicated, reordered and malformed cells reject. Every cell passes with a rational lower margin of at least

$$\frac{256695813}{10^{12}} > 0.$$

This proves Theorem 1 from the stated analytic inputs. The finite arithmetic is executable evidence for the displayed inequalities; it does not replace the count identity, comparison theorem or concavity arguments above.

5 Scope, checks and preserved failed shortcuts

The acceptance program is `compound-accept.py`; it checks 23 normalization gates and 8 malformed-input or source-change controls, reads the evaluator bytes, verifies their SHA256, and executes those exact bytes without importing cached bytecode. The finite exponential calculation is `compound-evaluator.py`. Its separate controls include zero intensities, exact hinge collisions,

finite positive-series lower bounds, and malformed floating/Boolean inputs after warming the exact cache. Floating scripts `compound-cover-discovery.py` and `compound-plan.py` selected the parameters but are not acceptance inputs. The 294 tested discovery boxes and the first conservative failed pilot are preserved separately from the exact run.

Two tempting shortcuts are false. First, the physical variance fraction cannot be reused as the own-count variance fraction. A centered actual law with $n = 42, p = 1/2, k = 22, r = .08$, $U_0 \in \{-.6, .001\}$ and $U_1 = -U_0$ has $v_0 = v_1 = .0006$, $w = 1/2$, but own variances $20v_0$ and $22v_1$. For $\kappa = -4r$ it violates the corresponding shortcut fourth-moment bound by $423903/2500000000$; its raw $|H| = .92/\sqrt{.2506} < 1.84$. Second, the centered compound hinge need not be convex in allocation: at $T = 1/40, w = 1/4, y = 1/5$ its second derivative is

$$-e^{-815/648}\frac{787398879114798488065}{147382613147155571933184} < 0.$$

The independent comparator review derives both controls exactly and supplies an independent probability-generating-function recurrence for (9). Neither false shortcut is used here or in the previous accepted small-noise theorem.

The new comparison can replace the negative-inner part of a selected right-panel low-noise branch, provided its strip, threshold and noise hypotheses have been acquired separately and the remaining phases are covered. It makes half-amplitude confinement and subsequent count-curvature arguments unnecessary *within this domain*. It does not justify deleting their uses in other probability panels, larger noise, or other acquisition and exclusion arguments. No manuscript or inherited certificate has been changed by this research component.